

Week 07: Inferential Statistics

Independent and paired t-tests, effect sizes, and boxplots

This report analyzes the earnings and screen time dataset.

Requirements:

- compare mean earnings
- compare paired sleep measurements
- visualize both comparisons
- reflect on statistical significance, effect sizes, and boxplots.

1. Dataset and analysis plan

The dataset contains 50 people and 5 columns. The analysis variables are earnings (in thousands of dollars), kids (0 = without kids, 1 = with kids), sleep_screen (minutes slept in the screen condition), and sleep_noscreen (minutes slept in the no-screen condition). The analysis variables contain no missing values.

A	B	C	D	E
Table1 ▾				
id ▾	earnings ▾	kids ▾	sleep_screen ▾	sleep_noscreen ▾
1	55	0	421	450
2	49	0	258	300
3	72	1	458	501
4	62	1	352	333
5	70	0	492	490

The earnings comparison includes 30 people with kids and 20 people without kids. These are separate groups, so Welch’s independent-samples t-test is used. The sleep comparison uses both measurements from each of 50 people, so a paired t-test is used. Tests are two-sided with $\alpha = 0.05$.

2. Independent t-test: Earnings

Research question and hypotheses

Research question: Do mean earnings differ between people without kids and people with kids?

$H_0: \mu_1 - \mu_0 = 0$ (the population means are equal). $H_1: \mu_1 - \mu_0 \neq 0$ (the population means differ).

Where μ_1 is population mean earnings for people without kids,

and μ_0 is population mean earnings for people with kids

Descriptive statistics:

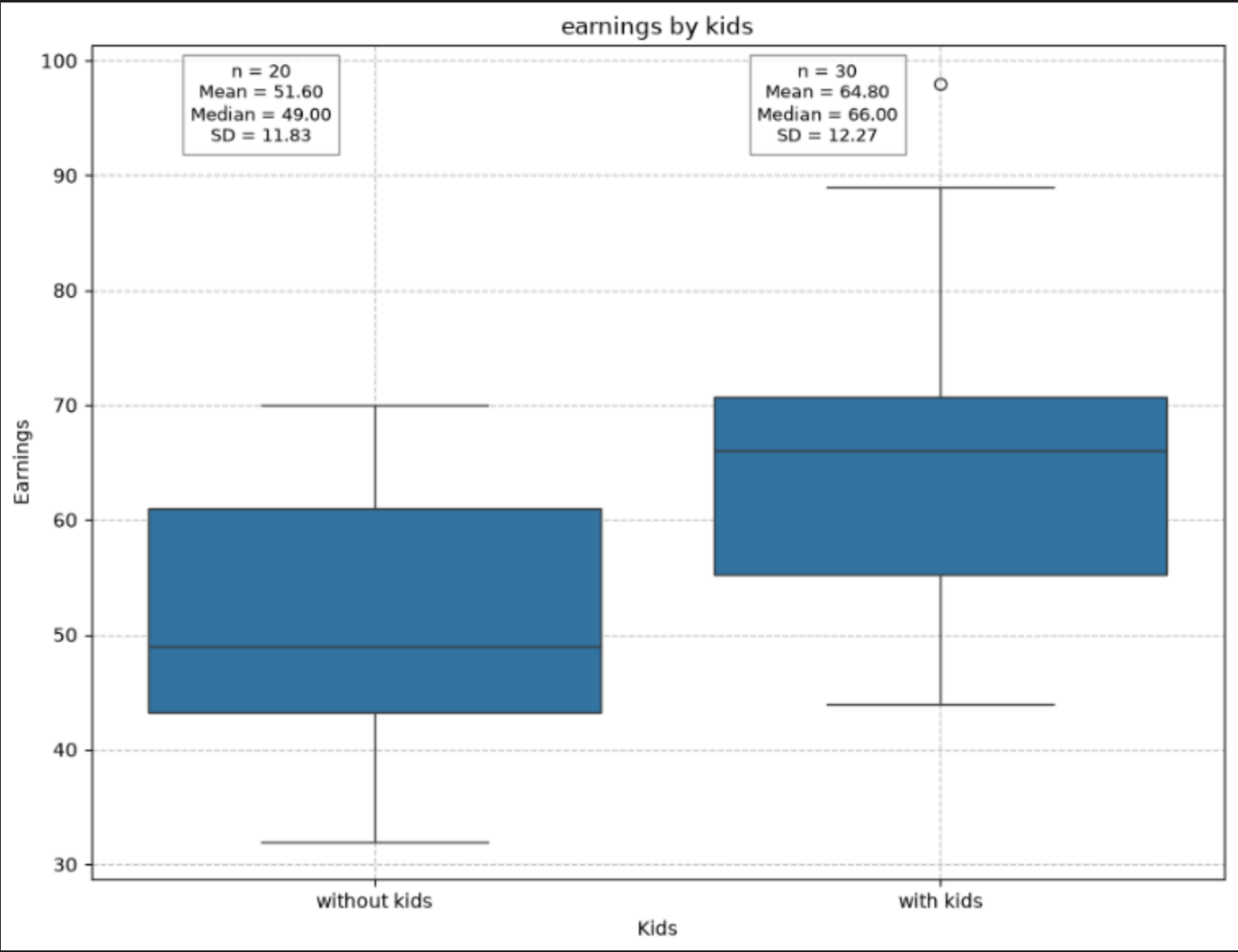
Group	Count (n)	Mean	SD
Without kids	20	51.60	11.83
With kids	30	64.80	12.27

Test Result

t-statistic	p-value	Effect size (r)	Significant at 0.05?
-3.808	0.000451	0.507	True

Welch’s test was statistically significant, $t = -3.808$, $p = 0.000451$. Since $p < .05$, reject H_0 . The estimated mean earnings difference (without kids - with kids) is -\$13.2k. Its 95% confidence interval is approximately [-\$20.2k, -\$6.2k]. The interval excludes zero, consistent with the significant test. The effect size $r = 0.507$ is large by common correlation-style benchmarks (about .10 small, .30 medium, .50 large). This is an association and does not establish that having kids causes higher earnings.

The boxplot below shows the median earnings are about \$49k for people without kids and \$66k for people with kids. The boxes and whiskers indicate the distributions’ spread. The distributions overlap, and the group with kids has a potential outlier on the higher end. The boxplot describes medians and spread, while the t-test tests the difference in population means.



3. Paired t-test: Sleep

Research question and hypotheses

Research question: Does mean sleep duration differ between the screen and no-screen conditions for the same people?

Define $d_i = \text{sleep_screen}_i - \text{sleep_noscreen}_i$.
 $H_0: \mu_d = 0$ (the population mean paired difference is zero).
 $H_1: \mu_d \neq 0$ (the population mean paired difference is not zero).
Descriptive statistics:

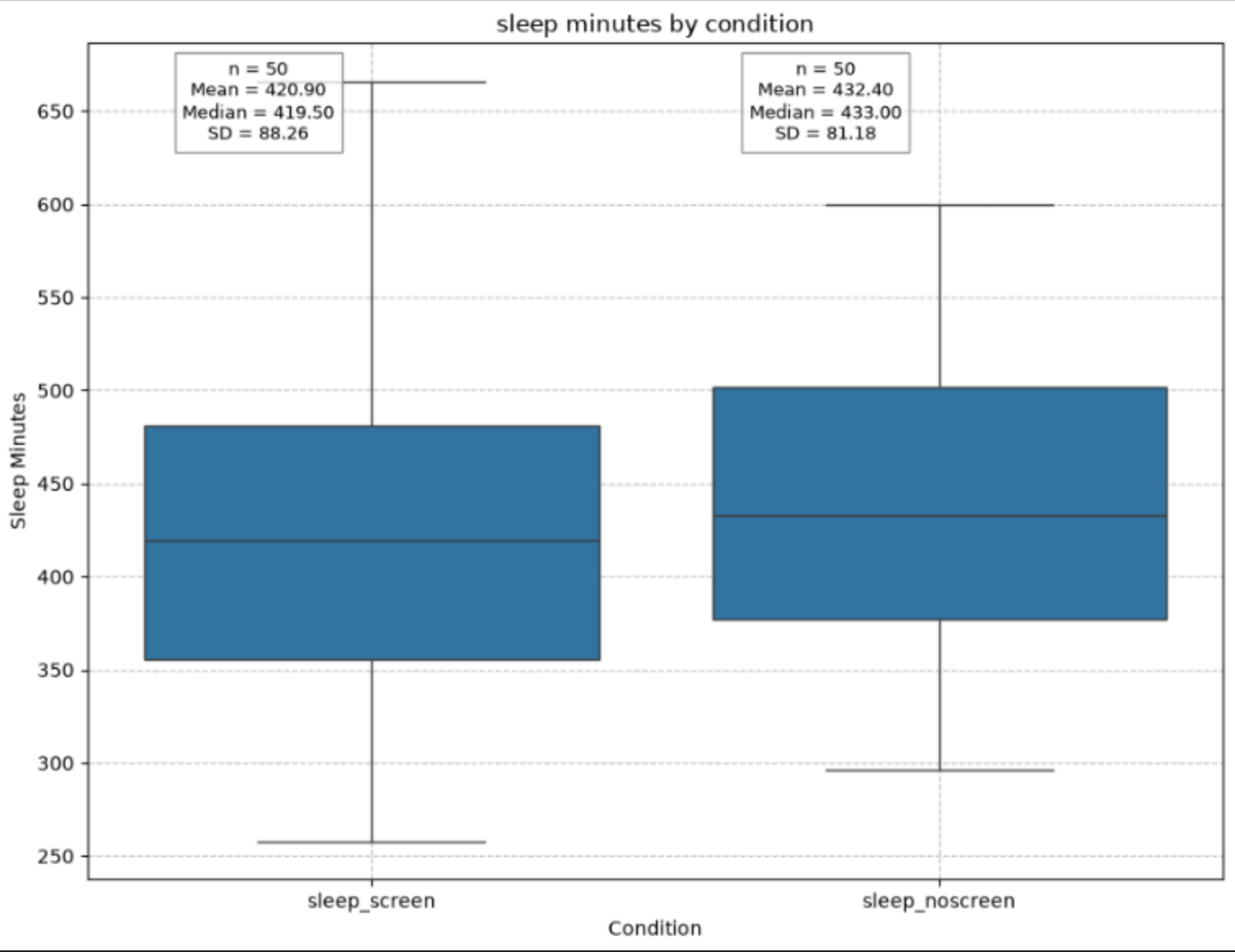
Condition	Mean	SD
Screen condition	420.9	88.26
No-screen condition	432.4	81.18

Test Result

t-statistic	p-value	Effect size (r)	Significant at 0.05?
-3.3894	0.001391	0.436	True

The paired t-test was statistically significant, $t = -3.389$, $p = 0.001391$. Since $p < .05$, reject H_0 . The mean paired difference (screen minus no screen) was -11.5 minutes, with a 95% confidence interval of [-18.32, -4.68] minutes. Because the interval is below zero, it is consistent with less sleep in the screen condition. The effect size $r = 0.436$ lies between the common medium (.30) and large (.50) benchmarks and is closer to large. The result alone does not establish causation.

The boxplot below shows the median is slightly higher in the no-screen condition (about 433 minutes) than in the screen condition (about 419.5 minutes). The distributions overlap considerably. The screen condition has a wider overall range in this sample. Since measurements are paired, separate boxplots do not display each participant’s change. The paired t-test analyzes within-person differences.



4. Overall summary

Both comparisons were statistically significant at $\alpha = .05$. For earnings, people without kids had lower sample mean earnings than people with kids by 13.2k dollars, Welch's $t = -3.808$, $p = 0.000451$, $r = 0.507$, 95% CI [-20.20, -6.20] k dollars. For sleep, the screen minus no-screen mean paired difference was -11.5 minutes, $t = -3.389$, $p = 0.001391$, $r = 0.436$, 95% CI [-18.32, -4.68] minutes.

The independent t-test is appropriate for separate groups while the paired t-test is appropriate for repeated measurements on the same people. The boxplots illustrate differences in medians and distributions, while the tests evaluate mean differences. Statistical significance does not imply causation. Effect-size cutoffs are guidelines and should be interpreted in context.